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O5 Solutions: algebra and inequalities

Read a solution only after a serious attempt. Final answers are in bold. For proof problems, compare the *structure* of your proof with the one here: did you prove every step, state the equality case, and (for functional equations) check the answer?

Level 1

Problem 1

Cube both sides: $\left(x+\frac{1}{x}\right)^3 = x^3+\frac{1}{x^3}+3\left(x+\frac{1}{x}\right)$. So $27 = x^3+\frac{1}{x^3}+9$ and $x^3+\frac{1}{x^3}=18$. (Such an x exists: $x=\frac{3\pm\sqrt{5}}{2}$.) Answer: **(B) 18**.

Problem 2

Put $a=2026$, $b=-2025$, $c=-1$. Then $a+b+c=0$, so $a^3+b^3+c^3=3abc$:

$$2026^3-2025^3-1 = 3\cdot 2026\cdot(-2025)\cdot(-1) = 3\cdot 2025\cdot 2026.$$

So the fraction equals 3. Answer: **(C) 3**.

(Without the identity: $2026^3-2025^3=(2025+1)^3-2025^3=3\cdot 2025^2+3\cdot 2025+1$, and subtracting 1 gives $3\cdot 2025\cdot 2026$.)

Problem 3

By the remainder theorem the remainder is $P(-1)=(-1)^{2030}+2\cdot(-1)+3=1-2+3=2$. Answer: **(C) 2**.

Problem 4

By Vieta $a+b=7$ and $ab=3$, so $a^2+b^2=(a+b)^2-2ab=49-6=43$. Answer: **(B) 43**.

Problem 5

For each k,

$$\frac{1}{\sqrt{k}+\sqrt{k+1}} = \frac{\sqrt{k+1}-\sqrt{k}}{(k+1)-k} = \sqrt{k+1}-\sqrt{k}.$$

The sum from $k=1$ to 99 telescopes to $\sqrt{100}-\sqrt{1}=9$. Answer: **(A) 9**.

Problem 6

From $a_3=a_1+2d=10$ and $a_{10}=a_1+9d=31$ we get $7d=21$, so $d=3$ and $a_1=4$. Then $a_{20}=4+19\cdot 3=61$ and the sum is $\frac{20(4+61)}{2}=650$. Answer: **(C) 650**.

Problem 7

$\frac{x^2+9}{x} = x + \frac{9}{x} \geq 2\sqrt{x \cdot \frac{9}{x}} = 6$ by AM-GM, with equality when $x = \frac{9}{x}$, that is $x = 3$. So the minimum 6 is actually reached. Answer: **(B) 6**.

Problem 8

With $x = 2$: $f(2) + 2f\left(\frac{1}{2}\right) = 6$. With $x = \frac{1}{2}$: $f\left(\frac{1}{2}\right) + 2f(2) = \frac{3}{2}$. Multiply the second equation by 2 and subtract the first: $3f(2) = 3 - 6 = -3$, so $f(2) = -1$. (A function with this property exists: solving the same system for general x gives $f(x) = \frac{2}{x} - x$, and indeed $\frac{2}{x} - x + 2\left(2x - \frac{1}{x}\right) = 3x$.) Answer: **(A)** -1.

Level 2

Problem 9

By Sophie Germain (with $b = 1$), or by completing the square,

$$n^4 + 4 = (n^2 + 2)^2 - (2n)^2 = (n^2 - 2n + 2)(n^2 + 2n + 2).$$

For $n = 1$ we get 5, which is prime. For $n \geq 2$ the smaller factor is $n^2 - 2n + 2 = (n-1)^2 + 1 \geq 2$, and the larger factor is bigger still, so $n^4 + 4$ is a product of two integers that are both at least 2, hence not prime. Answer: $n = 1$ **only**.

Problem 10

Let $e_1 = a + b + c = 6$, $e_2 = ab + bc + ca$, $e_3 = abc$.

From $a^2 + b^2 + c^2 = e_1^2 - 2e_2$: $14 = 36 - 2e_2$, so $e_2 = 11$.

The three-cube identity gives $a^3 + b^3 + c^3 - 3abc = e_1((a^2 + b^2 + c^2) - e_2)$, so $36 - 3e_3 = 6 \cdot (14 - 11) = 18$, and $e_3 = 6$.

For the fourth powers use Newton's sums, $p_4 = e_1 p_3 - e_2 p_2 + e_3 p_1$:

$$p_4 = 6 \cdot 36 - 11 \cdot 14 + 6 \cdot 6 = 216 - 154 + 36 = 98.$$

Answer: $abc = 6$ **and** $a^4 + b^4 + c^4 = 98$.

Check: a, b, c are the roots of $x^3 - 6x^2 + 11x - 6 = (x-1)(x-2)(x-3)$, so $\{a, b, c\} = \{1, 2, 3\}$, and indeed $1 + 16 + 81 = 98$. (This also shows that real numbers with the given properties exist.)

Problem 11

Suppose the roots r, s are integers. By Vieta $r + s = -a$ and $rs = 2a = -2(r+s)$. So $rs + 2r + 2s = 0$, and adding 4:

$$(r + 2)(s + 2) = 4.$$

The integer factorisations of 4 (up to order) are $1 \cdot 4$, $2 \cdot 2$, $(-1) \cdot (-4)$ and $(-2) \cdot (-2)$. They give

$\{r, s\}$	$a = -(r+s)$	equation
$\{-1, 2\}$	-1	$x^2 - x - 2 = (x+1)(x-2)$
$\{0, 0\}$	0	$x^2 = 0$
$\{-3, -6\}$	9	$x^2 + 9x + 18 = (x+3)(x+6)$
$\{-4, -4\}$	8	$x^2 + 8x + 16 = (x+4)^2$

The last column checks that each value of a really works. Answer: $a \in \{-1, 0, 8, 9\}$.

Problem 12

Division by a polynomial of degree 2 leaves a remainder of degree at most 1: $P(x) = (x-1)(x-2)Q(x) + ax + b$. By the remainder theorem $P(1) = 3$ and $P(2) = 5$, so $a + b = 3$ and $2a + b = 5$. Hence $a = 2$, $b = 1$. Answer: $2x + 1$. (For example $P(x) = 2x + 1$ itself satisfies the conditions, so the situation is possible.)

Problem 13

(a) Let $b_n = a_n - 1$. Then $b_{n+1} = a_n^2 - 2a_n + 1 = (a_n - 1)^2 = b_n^2$, and $b_1 = 2$. By induction $b_n = 2^{2^{n-1}}$: true for $n = 1$, and if $b_n = 2^{2^{n-1}}$ then $b_{n+1} = 2^{2 \cdot 2^{n-1}} = 2^{2^n}$. So $a_n = 2^{2^{n-1}} + 1$. (These are the Fermat numbers $3, 5, 17, 257, \ldots$)

(b) Induction on n. For $n = 1$: $a_1 = 3 = 5 - 2 = a_2 - 2$. Suppose $a_1 \cdots a_n = a_{n+1} - 2$. Then

$$a_{n+2} - 2 = a_{n+1}^2 - 2a_{n+1} = a_{n+1}(a_{n+1} - 2) = a_{n+1} \cdot a_1 \cdots a_n,$$

which is the statement for $n + 1$. □

(c) Let $m < n$ and let d be a positive common divisor of a_m and a_n. By (b), $a_n - 2 = a_1 \cdots a_{n-1}$ is a multiple of a_m, so $d \mid a_n - 2$. Together with $d \mid a_n$ this gives $d \mid 2$. But a_n is odd by (a), so $d = 1$. Hence $\gcd(a_m, a_n) = 1$. □

Problem 14

Since

$$\frac{1}{k(k+1)} - \frac{1}{(k+1)(k+2)} = \frac{(k+2) - k}{k(k+1)(k+2)} = \frac{2}{k(k+1)(k+2)},$$

each term is $\frac{1}{2}\left(\frac{1}{k(k+1)} - \frac{1}{(k+1)(k+2)}\right)$, and the sum telescopes:

$$\sum_{k=1}^{n} \frac{1}{k(k+1)(k+2)} = \frac{1}{2}\left(\frac{1}{2} - \frac{1}{(n+1)(n+2)}\right) = \frac{(n+1)(n+2) - 2}{4(n+1)(n+2)}.$$

Since $(n+1)(n+2) - 2 = n^2 + 3n$, the answer is $\dfrac{\mathbf{n(n+3)}}{\mathbf{4(n+1)(n+2)}}$. Check for $n = 1$: $\frac{4}{24} = \frac{1}{6}$. Correct.

Problem 15

By AM-GM, $a+b \geq 2\sqrt{ab}$, $b+c \geq 2\sqrt{bc}$ and $c+a \geq 2\sqrt{ca}$. All six sides are positive, so we may multiply:

$$(a+b)(b+c)(c+a) \geq 8\sqrt{a^2b^2c^2} = 8abc.$$

Equality in a product of inequalities between positive numbers holds only if each of them is an equality, that is $a=b$, $b=c$ and $c=a$. So **equality holds if and only if** $a=b=c$. □

Problem 16

By the Engel form of Cauchy–Schwarz,

$$\frac{1^2}{x} + \frac{2^2}{y} + \frac{3^2}{z} \geq \frac{(1+2+3)^2}{x+y+z} = \frac{36}{1} = 36.$$

Equality holds if and only if $\frac{1}{x} = \frac{2}{y} = \frac{3}{z}$, so $y = 2x$ and $z = 3x$. With $x+y+z = 1$ this gives $6x = 1$. **Equality holds only for** $(x,y,z) = \left(\frac{1}{6}, \frac{1}{3}, \frac{1}{2}\right)$, and indeed $6 + 12 + 18 = 36$. □

Problem 17

Write each fraction as $\frac{a^2}{a(b+c)}$ and apply the Engel form:

$$\frac{a^2}{a(b+c)} + \frac{b^2}{b(c+a)} + \frac{c^2}{c(a+b)} \geq \frac{(a+b+c)^2}{2(ab+bc+ca)}.$$

Now $(a+b+c)^2 = a^2+b^2+c^2+2(ab+bc+ca) \geq 3(ab+bc+ca)$, because $a^2+b^2+c^2 \geq ab+bc+ca$ (sum of the three inequalities $\frac{1}{2}(a-b)^2 \geq 0$ and so on). Hence the right side is at least $\frac{3(ab+bc+ca)}{2(ab+bc+ca)} = \frac{3}{2}$. □

Equality. The second step needs $a=b=c$. Conversely, if $a=b=c$ every fraction equals $\frac{1}{2}$ and the sum is $\frac{3}{2}$. So **equality holds if and only if** $a=b=c$.

Problem 18

By AM-GM on the three numbers a^3, a^3, b^3:

$$\frac{a^3+a^3+b^3}{3} \geq \sqrt[3]{a^6b^3} = a^2b.$$

In the same way $\frac{2b^3+c^3}{3} \geq b^2c$ and $\frac{2c^3+a^3}{3} \geq c^2a$. Adding the three gives $a^3+b^3+c^3 \geq a^2b + b^2c + c^2a$. □

Equality needs equality in all three AM-GM steps: $a=b$, $b=c$, $c=a$. So **equality holds if and only if** $a=b=c$.

Alternative (rearrangement). The triples (a^2,b^2,c^2) and (a,b,c) are ordered the same way, so $a^2\cdot a + b^2\cdot b + c^2\cdot c$ is the largest of all sums $a^2\cdot u + b^2\cdot v + c^2\cdot w$ where (u,v,w) is a permutation of (a,b,c). Taking $(u,v,w) = (b,c,a)$ gives the inequality. (The equality case is easier to read off from the AM-GM proof.)

Problem 19

By Cauchy–Schwarz,

$$(x^2+y^2)(3^2+4^2) \geq (3x+4y)^2 = 100,$$

so $x^2+y^2 \geq 4$. Equality holds if and only if $(x,y) = t(3,4)$ for some t; then $9t+16t=10$, so $t=\frac{2}{5}$ and $(x,y) = \left(\frac{6}{5}, \frac{8}{5}\right)$. Check: $\frac{36+64}{25} = 4$. Answer: **minimum** 4**, reached only at** $(x,y) = \left(\frac{6}{5}, \frac{8}{5}\right)$.

(Geometrically: x^2+y^2 is the squared distance from the origin to a point on the line $3x+4y=10$, and that distance is at least $\frac{10}{5} = 2$.)

Problem 20

Suppose f is a solution. Replacing x by $1-x$ gives a second equation. For each fixed x we have

$$f(x) + 2f(1-x) = x^2,$$
$$2f(x) + f(1-x) = (1-x)^2.$$

Subtract the first from twice the second: $3f(x) = 2(1-x)^2 - x^2 = x^2 - 4x + 2$. So the only candidate is

$$f(x) = \frac{x^2-4x+2}{3}.$$

Check: $f(1-x) = \frac{(1-x)^2-4(1-x)+2}{3} = \frac{x^2+2x-1}{3}$, so $f(x)+2f(1-x) = \frac{x^2-4x+2+2x^2+4x-2}{3} = x^2$. Correct. Answer: $f(x) = \dfrac{x^2-4x+2}{3}$.

Problem 21

Put $x=y=0$: $f(0)^2 - f(0) = 0$, so $f(0)=0$ or $f(0)=1$.

Put $y=0$: $f(x)f(0) - f(0) = x$ for all x.

- If $f(0)=0$, this says $0=x$ for all x, which is false (take $x=1$).
- So $f(0)=1$, and then $f(x)-1=x$, that is $f(x)=x+1$.

Check: $(x+1)(y+1) - (xy+1) = x+y$. Correct. Answer: $f(x) = x+1$.

Level 3

Problem 22

Let $s = a+b+c$ and $q = ab+bc+ca$. Then $a^2+b^2+c^2 = s^2-2q$, and the three-cube identity gives

$$n = a^3+b^3+c^3-3abc = s\,(a^2+b^2+c^2-q) = s\,(s^2-3q).$$

Necessity. Suppose $3 \mid n$. If $3 \nmid s$, then $3 \nmid s^2$, so $3 \nmid s^2-3q$ and $3 \nmid n$, a contradiction. So $3 \mid s$. Then also $3 \mid s^2-3q$, so $9 \mid n$. Hence no n with $n \equiv 3$ or $n \equiv 6 \pmod 9$ can be written in this form.

Sufficiency. For every integer k (these identities can be checked by expanding, or with the formula $n = s(s^2 - 3q)$):

$$\begin{aligned}(a,b,c) = (k,k,k+1): &\quad s = 3k+1,\ s^2-3q = 1,\ n = 3k+1,\\ (a,b,c) = (k,k+1,k+1): &\quad s = 3k+2,\ s^2-3q = 1,\ n = 3k+2,\\ (a,b,c) = (k-1,k,k+1): &\quad s = 3k,\ s^2-3q = 3,\ n = 9k.\end{aligned}$$

(For the first: $q = k^2 + 2k(k+1) = 3k^2 + 2k$, and $s^2 - 3q = 9k^2 + 6k + 1 - 9k^2 - 6k = 1$. The others are similar.) So every $n \equiv 1, 2 \pmod 3$ and every multiple of 9 is reachable.

Answer: **exactly the integers n with $n \not\equiv 3$ and $n \not\equiv 6 \pmod 9$**, that is, the integers that are not divisible by 3 together with the multiples of 9.

Problem 23

Method 1 (new polynomial). Note that 1 is not a root ($1 - 1 - 1 = -1 \neq 0$), so all fractions are defined. For a root x, let $y = \frac{1+x}{1-x}$. Then $y \neq -1$ (otherwise $1 + x = x - 1$) and $x = \frac{y-1}{y+1}$. Substituting into $x^3 - x - 1 = 0$ and multiplying by $(y+1)^3$:

$$(y-1)^3 - (y-1)(y+1)^2 - (y+1)^3 = 0.$$

Expanding: $(y^3 - 3y^2 + 3y - 1) - (y^3 + y^2 - y - 1) - (y^3 + 3y^2 + 3y + 1) = -y^3 - 7y^2 + y - 1$. So each y satisfies

$$y^3 + 7y^2 - y + 1 = 0.$$

The three roots a, b, c are distinct (the discriminant of $x^3 - x - 1$ is $-4 \cdot (-1)^3 - 27 \cdot 1^2 = -23 \neq 0$), and the map $x \mapsto \frac{1+x}{1-x}$ is injective, so the three values of y are distinct and are exactly the three roots of this cubic. By Vieta their sum is -7.

Method 2 (logarithmic derivative). Since $\frac{1+x}{1-x} = -1 + \frac{2}{1-x}$, the sum is $-3 + 2\sum \frac{1}{1-a}$. For $P(x) = (x-a)(x-b)(x-c)$ we have $\frac{P'(x)}{P(x)} = \frac{1}{x-a} + \frac{1}{x-b} + \frac{1}{x-c}$, so $\sum \frac{1}{1-a} = \frac{P'(1)}{P(1)} = \frac{3-1}{-1} = -2$. The sum is $-3 - 4 = -7$.

Answer: -7.

Problem 24

By AM-GM, $a + b + c \geq 3\sqrt[3]{abc} = 3$. By QM-AM (or Cauchy–Schwarz), $a^2 + b^2 + c^2 \geq \frac{(a+b+c)^2}{3}$. Therefore

$$a^2 + b^2 + c^2 \geq (a+b+c) \cdot \frac{a+b+c}{3} \geq (a+b+c) \cdot 1 = a + b + c.$$

□

Equality. The QM-AM step needs $a = b = c$, and with $abc = 1$ this means $a = b = c = 1$. Conversely $a = b = c = 1$ gives $3 = 3$. So **equality holds if and only if** $a = b = c = 1$.

Note the idea: the inequality is not homogeneous (degree 2 against degree 1), and the condition $abc = 1$ supplies the missing factor $\frac{a+b+c}{3} \geq 1$.

Problem 25

Let $f(x) = \frac{x}{1+x^2}$. Equality should hold at $a = b = c = \frac{1}{3}$, where $f\left(\frac{1}{3}\right) = \frac{1/3}{10/9} = \frac{3}{10}$. The derivative is $f'(x) = \frac{1-x^2}{(1+x^2)^2}$, so $f'\left(\frac{1}{3}\right) = \frac{8/9}{100/81} = \frac{18}{25}$, and the tangent line at $x = \frac{1}{3}$ is

$$\ell(x) = \frac{3}{10} + \frac{18}{25}\left(x - \frac{1}{3}\right) = \frac{36x+3}{50}.$$

Claim: $f(x) \le \ell(x)$ for all $x > 0$. Since $1 + x^2 > 0$, the claim is equivalent to $50x \le (36x + 3)(1 + x^2)$, that is

$$36x^3 + 3x^2 - 14x + 3 \ge 0.$$

The left side factors as $(3x-1)^2(4x+3)$ (expand to check: $(9x^2 - 6x + 1)(4x + 3) = 36x^3 + 3x^2 - 14x + 3$). For $x > 0$ both factors are nonnegative, so the claim holds, with equality only at $x = \frac{1}{3}$.

Adding the claim for a, b and c:

$$\sum \frac{a}{1+a^2} \le \frac{36(a+b+c)+9}{50} = \frac{45}{50} = \frac{9}{10}.$$

□ **Equality holds if and only if** $a = b = c = \frac{1}{3}$**.**

Problem 26

Substitute $x = \frac{1}{a}$, $y = \frac{1}{b}$, $z = \frac{1}{c}$; these are positive and $xyz = 1$. Then

$$\frac{1}{a^3(b+c)} = \frac{x^3}{\frac{1}{y} + \frac{1}{z}} = \frac{x^3 yz}{y+z} = \frac{x^2 \cdot xyz}{y+z} = \frac{x^2}{y+z},$$

and similarly for the other terms. By the Engel form and then AM-GM,

$$\frac{x^2}{y+z} + \frac{y^2}{z+x} + \frac{z^2}{x+y} \ge \frac{(x+y+z)^2}{2(x+y+z)} = \frac{x+y+z}{2} \ge \frac{3\sqrt[3]{xyz}}{2} = \frac{3}{2}.$$

□

Equality. The AM-GM step needs $x = y = z$, so $x = y = z = 1$; then each term is $\frac{1}{2}$. So **equality holds if and only if** $a = b = c = 1$. (This is a classic IMO problem from 1995, restated.)

Problem 27

By induction all a_n are positive. Squaring the recurrence,

$$a_{k+1}^2 = a_k^2 + 2 + \frac{1}{a_k^2}.$$

Adding this for $k = 1, \dots, n-1$ gives

$$a_n^2 = 1 + 2(n-1) + \sum_{k=1}^{n-1} \frac{1}{a_k^2} = 2n - 1 + \sum_{k=1}^{n-1} \frac{1}{a_k^2}.$$

In particular $a_n^2 \geq 2n - 1$ for every $n \geq 1$.

Lower bound. $a_{100}^2 = 199 + S$ with $S = \sum_{k=1}^{99} \frac{1}{a_k^2} > 0$. So $a_{100}^2 > 199 > 196$, and since $a_{100} > 0$, $a_{100} > 14$.

Upper bound. Using $a_k^2 \geq 2k - 1 \geq k$, we get $S \leq \sum_{k=1}^{99} \frac{1}{k}$. Split $1, 2, \ldots, 127$ into the seven blocks $[2^j, 2^{j+1} - 1]$ for $j = 0, 1, \ldots, 6$. Block j has 2^j terms, each at most $\frac{1}{2^j}$, so its sum is at most 1, and for $j \geq 1$ it is strictly less than 1. Hence

$$S \leq \sum_{k=1}^{99} \frac{1}{k} \leq \sum_{k=1}^{127} \frac{1}{k} < 7,$$

so $a_{100}^2 < 199 + 7 = 206 < 225$ and $a_{100} < 15$. □

(Numerically $a_{100} \approx 14.21$.)

Problem 28

Answer: $P = 0$ **and** $P(x) = x^n$ **for** $n = 0, 1, 2, \ldots$ (so $n = 0$ gives the constant 1).

Check: $0 = 0^2$, and $(x^2)^n = (x^n)^2$.

These are the only ones. Let $P \neq 0$. Two polynomials that agree for all real x have the same coefficients, so we compare coefficients.

Case 1: P has one nonzero term, $P(x) = cx^n$ with $c \neq 0$. Then $cx^{2n} = c^2 x^{2n}$, so $c = c^2$ and $c = 1$.

Case 2: P has at least two nonzero terms. Let $n = \deg P$ with leading coefficient $c_n \neq 0$, and let $k < n$ be the largest index below n with $c_k \neq 0$. So $P(x) = c_n x^n + c_k x^k + R(x)$ with $\deg R < k$ (or $R = 0$). Then

$$P(x)^2 = c_n^2 x^{2n} + 2c_n c_k x^{n+k} + \big(\text{terms of degree} < n + k\big),$$

because the remaining products $c_k^2 x^{2k}$, $2c_n x^n R(x)$, $2c_k x^k R(x)$ and $R(x)^2$ all have degree less than $n + k$. So the coefficient of x^{n+k} in $P(x)^2$ is $2c_n c_k \neq 0$.

On the other side, $P(x^2) = \sum_j c_j x^{2j}$. If $n + k$ is odd, x^{n+k} does not appear at all. If $n + k$ is even, its coefficient is $c_{(n+k)/2}$, and $k < \frac{n+k}{2} < n$, so this coefficient is 0 by the choice of k. In both cases the coefficients of x^{n+k} differ, a contradiction. So Case 2 is impossible. □

Problem 29

Let $g(x) = f(x) - \frac{x^2}{2}$, a function $\mathbb{Q} \to \mathbb{Q}$. Then

$$g(x + y) = f(x) + f(y) + xy - \frac{x^2 + 2xy + y^2}{2} = g(x) + g(y),$$

so g satisfies Cauchy's equation on $\mathbb{Q}$.

Cauchy over $\mathbb{Q}$ (proof from the notes): $x = y = 0$ gives $g(0) = 0$; induction gives $g(mx) = mg(x)$ for positive integers m; $g(x) + g(-x) = g(0) = 0$ extends this to all integers m; and for $x = \frac{m}{n}$ with $n > 0$, $ng(\frac{m}{n}) = g(m) = mg(1)$. So $g(x) = cx$ with $c = g(1) = f(1) - \frac{1}{2}$, which is rational because $f(1)$ is.

So $f(x) = \frac{x^2}{2} + cx$ for some rational c.

Check: for any rational c, f maps $\mathbb{Q}$ to $\mathbb{Q}$, and

$$f(x+y) = \frac{(x+y)^2}{2} + c(x+y) = \left(\frac{x^2}{2} + cx\right) + \left(\frac{y^2}{2} + cy\right) + xy = f(x) + f(y) + xy.$$

Answer: $f(x) = \dfrac{x^2}{2} + cx$**, where** c **is any rational number**.

Problem 30

Answer: $f(x) = x$**.**

Check: $x^2 + y = y + x^2$. Correct.

Uniqueness. Write $c = f(0)$.

Step 1: f **is bijective.** Put $x = 0$: $f(f(y)) = y + c^2$ for all y. If $f(y_1) = f(y_2)$, then $y_1 + c^2 = f(f(y_1)) = f(f(y_2)) = y_2 + c^2$, so $y_1 = y_2$: f is injective. For any real t, $f(f(t - c^2)) = t$: f is surjective.

Step 2: $f(0) = 0$**.** Comparing the equation for x and for $-x$ (same left side, since $x^2 = (-x)^2$) gives $f(x)^2 = f(-x)^2$ for all x. By surjectivity there is an a with $f(a) = 0$. Then $f(-a)^2 = f(a)^2 = 0$, so $f(-a) = 0 = f(a)$, and by injectivity $-a = a$, so $a = 0$. Thus $f(0) = 0$ and $c = 0$.

Step 3: two consequences. Now Step 1 reads $f(f(y)) = y$. Putting $y = 0$ in the equation gives $f(x^2) = f(x)^2$. Hence for every $t \geq 0$, $f(t) = f(\sqrt{t})^2 \geq 0$.

Step 4: f **is non-decreasing.** For $t \geq 0$ put $x = \sqrt{t}$: $f(t + f(y)) = y + f(\sqrt{t})^2 = y + f(t)$. Now put $y = f(z)$ and use $f(f(z)) = z$:

$$f(t + z) = f(z) + f(t) \qquad \text{for all } t \geq 0 \text{ and all real } z.$$

If $u \geq v$, take $t = u - v \geq 0$ and $z = v$: $f(u) = f(v) + f(u - v) \geq f(v)$.

Step 5: $f(x) = x$**.** Suppose $f(x) > x$ for some x. Since f is non-decreasing, $f(f(x)) \geq f(x) > x$, but $f(f(x)) = x$. Contradiction. If $f(x) < x$, then $f(f(x)) \leq f(x) < x$, again a contradiction. So $f(x) = x$ for all x. □

(Notice that we never needed to solve Cauchy's equation over $\mathbb{R}$: monotonicity together with $f(f(x)) = x$ finished the problem directly.)